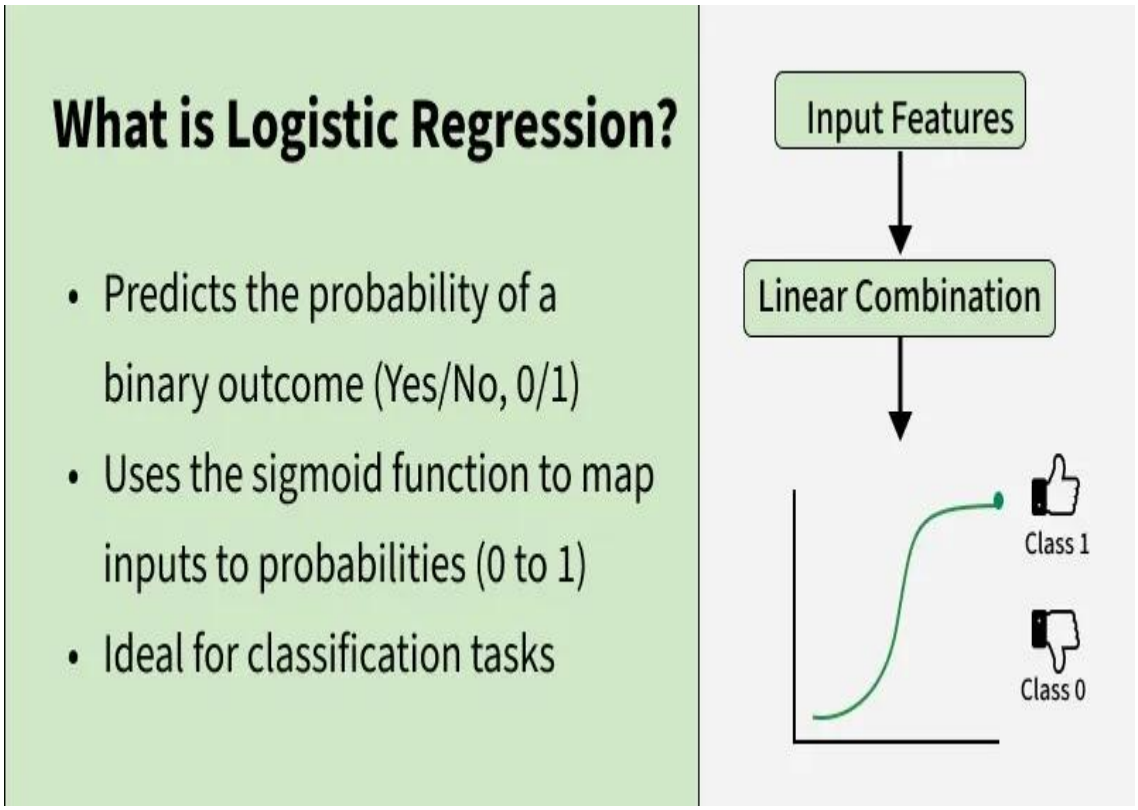
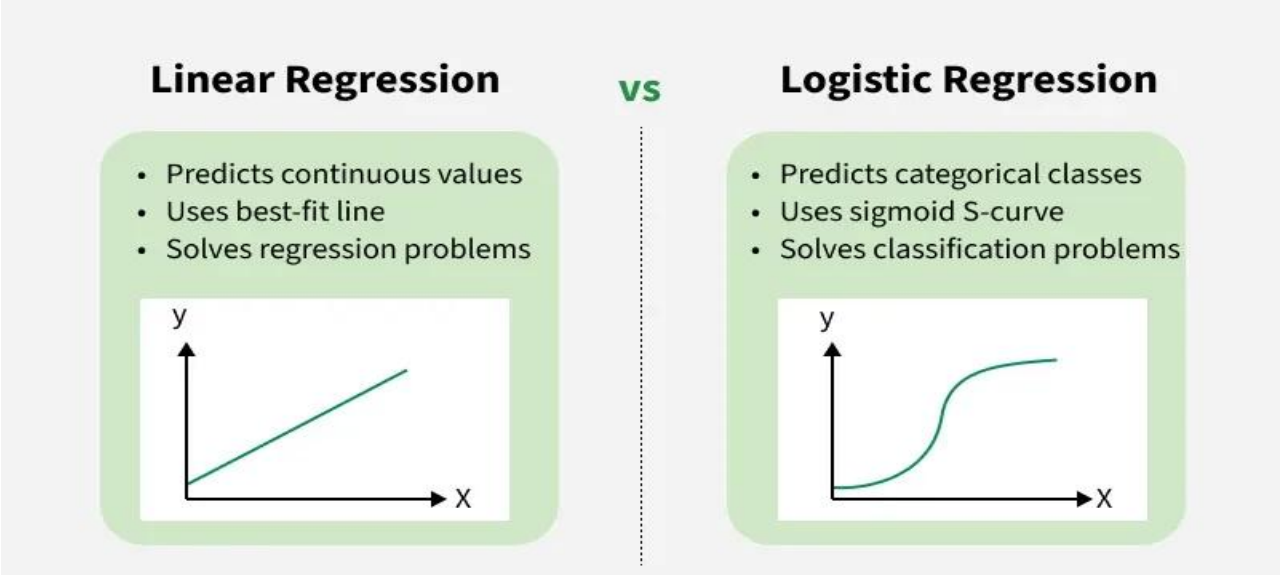
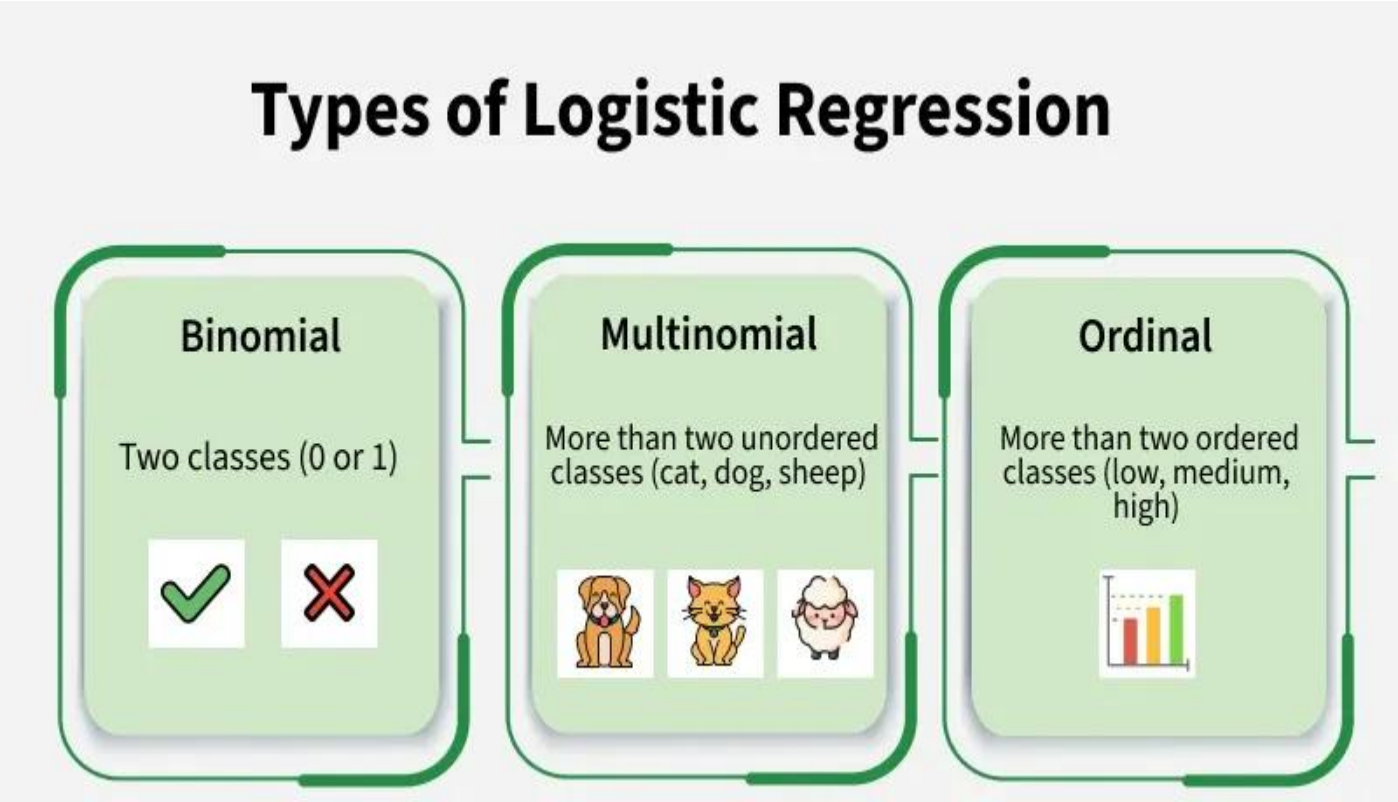
Logistic Regression in Machine Learning

Logistic Regression is a supervised machine learning algorithm used for classification problems. Unlike linear regression, which predicts continuous values it predicts the probability that an input belongs to a specific class.

- It is used for binary classification where the output can be one of two possible categories such as Yes/No, True/False or 0/1.
- It uses sigmoid function to convert inputs into a probability value between 0 and 1.







Types

1. **Binomial Logistic Regression:** Used when the dependent variable has only two possible categories, such as Yes/No, Pass/Fail, or 0/1. It is the most common type and is used for binary classification tasks.
2. **Multinomial Logistic Regression:** Used when the dependent variable has three or more unordered categories. For example, classifying animals as cat, dog, or sheep. It extends logistic regression to handle multiple classes.
3. **Ordinal Logistic Regression:** Used when the dependent variable has three or more categories with a natural order, such as Low, Medium, and High. It considers the ranking of categories during prediction.

Assumptions

1. **Independent Observations:** Each data point should be independent of the others, meaning there should be no dependence between observations.
2. **Binary Dependent Variable:** The target variable is typically binary and can take only two values, such as 0/1 or Yes/No. For multiclass problems, extensions such as Softmax-based logistic regression are used.

3. **Linearity of Log-Odds:** The independent variables should have a linear relationship with the log-odds of the dependent variable.
4. **No Extreme Outliers:** Extreme outliers can distort coefficient estimates and negatively affect model performance.
5. **Large Sample Size:** A sufficiently large dataset helps produce more reliable and stable predictions.

Understanding Sigmoid Function

1. The sigmoid function is a key component of Logistic Regression that converts the model's raw output into a probability value between 0 and 1.

2. This function takes any real number and maps it into the range 0 to 1 forming an "S" shaped curve called the sigmoid curve or logistic curve. Because probabilities must lie between 0 and 1, the sigmoid function is perfect for this purpose.

3. In logistic regression, we use a threshold value usually 0.5 to decide the class label.

- If the sigmoid output is same or above the threshold, the input is classified as Class 1.
- If it is below the threshold, the input is classified as Class 0.

This approach helps to transform continuous input values into meaningful class predictions.

Differences Between Linear and Logistic Regression

Aspect	Linear Regression	Logistic Regression
Definition	Linear regression is used to predict the continuous dependent variable using a given set of independent variables.	Logistic regression is used to predict the categorical dependent variable using a given set of independent variables.
Problem Type	It is used for solving regression problem.	It is used for solving classification problems.
Output Type	In this we predict the value of continuous variables.	In this we predict values of categorical variables.
Curve/Model Fitting	In this we find best fit line.	In this we find S-Curve.

Aspect	Linear Regression	Logistic Regression
Estimation Method	Least square estimation method is used for estimation of accuracy.	Maximum likelihood estimation method is used for estimation of accuracy.
Output Example	The output must be continuous value such as price, age etc.	Output must be categorical value such as 0 or 1, Yes or No, etc.

Terminologies Used

1. **Independent Variables:** These are the input features or predictor variables used to make predictions about the dependent variable.
2. **Dependent Variable**: This is the target variable that we aim to predict. In logistic regression, the dependent variable is categorical.
3. **Logistic Function**: This function transforms the independent variables into a probability between 0 and 1 which represents the likelihood that the dependent variable is either 0 or 1.
4. **Odds**: This is the ratio of the probability of an event happening to the probability of it not happening. It differs from probability because probability is the ratio of occurrences to total possibilities.
5. **Log-Odds (Logit)**: The natural logarithm of the odds. In logistic regression, the log-odds are modeled as a linear combination of the independent variables and the intercept.
6. **Coefficient**: These are the parameters estimated by the logistic regression model which shows how strongly the independent variables affect the dependent variable.
7. **Intercept**: The constant term in the logistic regression model which represents the log-odds when all independent variables are equal to zero.
8. **Maximum Likelihood Estimation (MLE)**: This method is used to estimate the coefficients of the logistic regression model by maximizing the likelihood of observing the given data.